

Measuring Earth's Magnetic Field with a Smart Device – Background Information and Additional Resources

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Background information:

- The Earth has a dipolar **magnetic field** that is mostly generated by the movement of iron in the Earth's liquid outer core (Olsen & Pauluhn, 2019).
- The Earth's magnetic field is a **vector** – it has a strength and a direction that we can measure (Olsen & Pauluhn, 2019).
- **Magnetic field intensity (F)** describes the strength of Earth's magnetic field and varies strongly by location. It is defined by three orthogonal components:

X: the north component

Y: the east component

Z: the vertical component

Intensity is generally measured in **nanoteslas** (nT) and varies between 25,000 – 65,000 nT (NCEI, 2025).

- **Magnetic inclination (I)** is the angle that the Earth's magnetic field lines make with the surface. It is measured in degrees (°) and ranges from $\pm 90^\circ$ at the poles to 0° at the equator (GFZ, 2025).
 - Inclination is positive in the northern hemisphere, where the field lines point down into the ground.
 - Inclination is negative in the southern hemisphere where the field lines point up out of the ground.
- The word magnetometer is a combination of the words **magnet + meter**.
- A **magnetometer** is any device capable of measuring the strength, direction or changes in Earth's magnetic field. A field compass is a simple magnetometer, that gives the direction of Earth's magnetic field (Ungvarsky, 2023).

- Smart devices like smartphones and tablets contain built-in **3-axis magnetometer sensors**. The devices use the magnetometer for navigation – when you’re locating yourself using Google Maps, you are using data from your phone’s magnetometer! (NASA, 2025).
- **Magnetic interference** is unwanted noise or interference due to contact with an external magnetic source (Wright, 2022). Interference can disrupt a magnetic sensor’s measurements and introduce errors in geomagnetic data.
- Sources of magnetic interference include cars, steel buildings, nearby electronic devices and other magnetic sources such as power lines, motors and air con units (U.S. EPA, 2016).

References:

GFZ German Research Centre for Geosciences (2025). *Frequently asked questions regarding geomagnetism*. Available at: <https://www.gfz.de/en/section/geomagnetism/data-products-services/frequently-asked-questions-regarding-geomagnetism#c14403>

NASA (2025). *Introductory Experiments in Magnetism with Smart Devices*. Available at: <https://science.nasa.gov/learn/heat/resource/introductory-experiments-in-magnetism-with-smart-devices/>

National Centers for Environmental Information (NCEI) (2025). *Geomagnetism frequently asked questions*. National Oceanic and Atmospheric Administration. Available at: <https://www.ncei.noaa.gov/products/geomagnetism-frequently-asked-questions>

Olsen, N. and Pauluhn, A. (2019). Exploring Earth’s magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at: https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

U.S. Environmental Protection Agency (EPA) (2016). *Magnetic methods*. Available at: https://archive.epa.gov/esd/archive-geophysics/web/html/magnetic_methods.html

Ungvarsky, J. (2023). *Magnetometer*, EBSCO Research Starters. Available at: <https://www.ebsco.com/research-starters/history/magnetometer>

Wright, G. (2022). *Electromagnetic interference (EMI)*. Available at: <https://www.techtarget.com/searchmobilecomputing/definition/electromagnetic-interference>

Additional resources:

For further information on the generation of the Earth's magnetic field and how it is studied: Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at:
https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

For additional classroom activities using smart device magnetometers:

NASA (2025). *Introductory Experiments in Magnetism with Smart Devices*. Available at:
<https://science.nasa.gov/learn/heat/resource/introductory-experiments-in-magnetism-with-smart-devices/>

NASA (2025). *Exploratory Activities in Magnetism with Smart Devices*. Available at:
<https://science.nasa.gov/learn/heat/resource/exploratory-experiments-in-magnetism-with-smart-devices/>